MIL-B-52706(ME)
27 April 1971
SUPERSEDING
MIL-P-52076(CE)
23 June 1959
MIL-C-52084(CE)
10 September 1959
MIL-P-52085(CE
15 September 1959

MILITARY SPECIFICATION

BRIDGE, FLOATING: RAFT SECTION, LIGHT TACTICAL,

SUPERSTRUCTURE COMPONENTS

1. SCOPE

1.1 Scope. This specification covers the panel and curb components comprising the superstructure for the raft section of the light tactical floating bridge.

1.2 Classification. The components shall be of the following types and classes, as specified:

Type I - Panel
  Class 1 - Deck
  Class 2 - Articulator
  Class 3 - Ramp, Female
  Class 4 - Ramp, Male
  Class 5 - Filler, Deck
  Class 6 - Filler, Articulator and Ramp
Type II - Curb
  Class 1 - Deck
  Class 2 - Articulator and Ramp

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

FSC 5420

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SPECIFICATIONS

Federal

QQ-S-781	- Strapping, Steel, Flat and Seals.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-C-650	- Crates, Wood, Open and Covered.

Military

MIL-T-704	- Treatment and Painting of Materiel.

STANDARDS

Federal

FED. STD. No. 595	- Colors.

Military

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.

DRAWINGS

ME

DL13216E9590	- Panel Assembly, Deck.
DL13216E9604	- Panel Assembly, Articulator.
DL13216E9613	- Panel Assembly, Ramp, Female.
DL13216E9615	- Panel, Ramp, Male.
DL13216E9616	- Panel, Filler, Deck Panel.
DL13216E9617	- Panel, Filler, Articulator and Ramp Panels.
DL13216E9618	- Curb Assembly, Deck Panel.
DL13216E9620	- Curb Assembly, Articulator and Ramp Panels.

(Copies of specifications, standards, and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

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AMERICAN WELDING SOCIETY

A5.1 - Mild Steel Covered Arc-Welding Electrodes.
A5.10 - Aluminum and Aluminum-Alloy Welding Rods and Bare Electrodes.
A5.18 - Mild Steel Electrodes for Gas Metal-Arc Welding.
B3.0 - Standard Qualification Procedure.
D2.0 - Welded Highway and Railway Bridges.

(Application for copies should be addressed to the American Welding Society, 345 East 47th Street, New York, N. Y. 10017.)

AMERICAN SOCIETY OF MECHANICAL ENGINEERS

Boiler and Pressure Vessel Code, Section IX, Welding Qualifications.

(Application for copies should be addressed to the American Society of Mechanical Engineers, 345 East 47th Street, New York, N. Y. 10017.)

UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classification

(Application for copies should be addressed to the Tariff Publishing Officer, Room 202, Union Station, 516 West Jackson Boulevard, Chicago, Illinois 60606.)

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Associations, Inc., ATTN: Tariff Order Section, 1616 P Street NW, Washington, D. C. 20036.)

3. REQUIREMENTS

3.1 Description. The panel and curb components shall be as shown on the drawings specified in 3.8 through 3.15, as applicable, and as specified herein.

3.1.1 Drawings. The drawings forming a part of this specification are engineering design drawings. No deviation from the prescribed dimensions or tolerances is permissible without prior approval of the contracting officer. Any data (e.g. shop drawings, layouts, flow sheets, processing

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procedures, etc.) prepared by the supplier or obtained from a vendor to support fabrication and manufacture of the production item shall be made available, upon request, for inspection by the contracting officer or his designated representative.

3.2 First article (preproduction model). The supplier shall furnish one or more of each type and class of components, as applicable, for inspection within the time frame specified to prove, prior to starting production, that his production methods will produce components that comply with the requirements of this specification. Examination and tests shall be as specified in section 4 and shall be subject to surveillance and approval by the Government (see 6.3).

3.3 Material. Material shall be as specified herein and as shown on the applicable drawings. Materials not specified shall be selected by the supplier and shall be subject to all provisions of this specification.

3.4 Welding electrodes.

3.4.1 For steel. Welding electrodes for steel shall conform to AWS A5.1 of A5.18, classification E60 or E70 series, size as required.

3.4.2 For aluminum. Welding electrodes for aluminum shall conform to AWS A5.10, classification ER4043, size as required.

3.5 Identification marking. The component shall be metal-die stamped as shown on the applicable drawing. Die stamping shall be legible after painting.

3.6 Treatment and painting. The component, except pins and threaded portions of threaded components and parts when applicable, shall be cleaned, treated, and painted in accordance with MIL-T-704, type B, except that the color number designation shall be 34087 in accordance with FED. STD. No. 595. Contact surfaces, regardless of whether such contact is between similar or dissimilar metals, shall be cleaned, treated, and painted prior to assembly.

3.7 Stenciling. The nomenclature, Federal stock number, weight, and cube shall be stenciled on each component. The letters "NET WT" shall precede the numerals in the weight and the letters "CU" shall precede the cubic displacement. Markings shall be Gothic-type capitals and Arabic numerals 3/4 inch high. Stenciling shall conform to MIL-T-704.

3.8 Type I, class 1, panel assembly, deck. The type I, class 1, deck panel assembly, shall be as shown on DL13216E9590 and as specified herein.

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3.9 Type I, class 2, panel assembly, articulator. The type I, class 2, articulator panel assembly, shall be as shown on DL13216E9604 and as specified herein.

3.10 Type I, class 3, panel assembly, ramp, female. The type I, class 3, female ramp panel assembly, shall be as shown on DL13216E9613 and as specified herein.

3.11 Type I, class 4, panel ramp, male. The type I, class 4, male ramp panel, shall be as shown on DL13216E9615 and as specified herein.

3.12 Type I, class 5, panel, filler, deck panel. The type I, class 5, deck filler panel, shall be as shown on DL13216E9696 and as specified herein.

3.13 Type I, class 6, panel, filler, articulator and ramp panels. The type I, class 6, articulator and ramp filler panel, shall be as shown on DL13216E9617 and as specified herein.

3.14 Type II, class 1, curb assembly, deck panel. The type II, class 1, deck panel curb assembly, shall be as shown on DL13216E9618 and as specified herein.

3.15 Type II, class 2, curb assembly, articulator and ramp panels. The type II, class 2, articulator and ramp panel curb assembly, shall be as shown on DL13216E9620 and as specified herein.

3.16 Workmanship.

3.16.1 Steel fabrication. Steel may be cut by shearing, sawing, or flame cutting. Burned surfaces of flame-cut material shall be ground or machined to remove cooling checks. All bends shall be made by use of metal dies or fixtures to insure uniformity in size and shape. All burrs shall be removed.

3.16.2 Aluminum fabrication. Aluminum used in the fabrication of the component shall be free from kinks and sharp bends. Straightening of material shall be done by methods that will not cause injury to the metal. Surfaces and edges of all finished members shall be smooth and free from kinks, bends, and winds. All bends of a major character shall be made with metal dies or fixtures to insure uniformity of size and shape. Sharp instruments shall not be used for marking bend lines. Aluminum shall not be cut by methods using heat. Aluminum less than 3/8 inch thick may be cut by shearing; aluminum 3/8 inch thick and over shall be cut by routing or by high-speed saws. Shearing and sawing shall be done neatly and accurately.

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3.16.3 Riveted connections.

3.16.3.1 Rivet holes. Holes for countersunk rivets shall be of such depth and angle that the rivet head shall fill the hole without projection or recession.

3.16.3.2 Riveting. Components shall be bolted in position before riveting. Jigs or frames shall be used to maintain alinement and tolerance. Rivets shall be free from slag or scale when driven. Countersunk rivets shall be flush with the surrounding surface. Heads of driven rivets shall be uniform in shape and concentric with the rivet shank axis, and shall have full bearing on the surfaces of the joined members. Rivets shall fill the holes in the joined members. Caulking or redriving of rivet heads shall not be permitted. Rivet tension shall not be created by driving in incorrectly fitted members. A rivet with a head deficient in size, not uniform in shape, off-center, loose, burned, or misshapen in any way shall be considered defective. Defective rivets shall be removed without injury to the joined metal. Automatic or squeeze-type riveters shall be permitted for driving rivets, where practicable, provided the pressure employed does not cause distortion of the material. In lieu of automatic or squeeze-type rivet machines, pneumatic rivet hammers may be used.

3.16.3.2.1 Steel rivets. Steel rivets may be driven hot or cold. Hot steel rivets shall be heated uniformly throughout their length and driven while at a light cherry color. Hot steel rivets in aluminum shall be driven at random in order to prevent overheating of the aluminum.

3.16.3.2.2 Aluminum rivets. Aluminum rivets may be driven hot or cold immediately after quenching. Aluminum rivets shall not be heated in excess of 950° F. and shall be driven at a temperature of not less than 850° F.

3.16.4 Welders and welding.

3.16.4.1 Welders. Before assigning any welder to manual welding work covered by this specification, the supplier shall provide the contracting officer with certification that the welder has passed qualification tests as prescribed by either of the following listed codes for the materials joined and the type of welding operation to be performed and that such qualification is effective as defined by the particular code:

Standard Qualification Procedure of the American Welding Society (B3.0).
Welding Qualifications of the ASME Boiler and Pressure Vessel Code (Section IX).

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3.16.4.2 Welding. The surfaces of parts to be welded shall be free from scale, paint, grease, and other foreign matter. Welding shall be done by the shielded metal-arc or gas metal-arc (MIG) process. Steel welding process shall conform to the provisions of the American Welding Society publication "D2.0 - Welded Highway and Railway Bridges". Aluminum welding process shall conform to the provisions of Part B, Section IX (Welding Qualifications) of the ASME Boiler and Pressure Vessel Code. All welded parts shall be free from cracks and other imperfections that may reduce the effectiveness of the part. Work shall be positioned for flat welding when practicable. Before welding over previously deposited weld metal, all traces of slag shall be removed and the deposit and adjoining base metal shall be wire brushed clean.

3.16.4.2.1 Steel welding. Where embrittlement of the joined metal may result, preheating or annealing or both shall be employed.

3.16.4.2.2 Aluminum welding. Aluminum members to be welded may be preheated for welding to a temperature of not more than 400° F. Aluminum welds may be filed, wire brushed, or chipped but shall not be hammered.

3.16.5 Interchangeability. Components of like description shall be fabricated to be duplicates within the tolerances specified and shall be interchangeable.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. The supplier is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards.

4.1.2 Parts and components. Parts and components detailed on the drawings shall be inspected in accordance with the Supplementary Quality Assurance Provisions (SQAP) shown on the drawings. The drawings specify the characteristics requiring SQAP inspection, the sampling plan, and the basis for acceptance and rejection (see 6.4 and 6.6).

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4.2 Classification of inspection. Inspection shall be classified as follows:

(a) Preproduction inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).
(c) Inspection of preparation for delivery (see 4.6).

4.3 Preproduction inspection.

4.3.1 Examination. The preproduction component shall be examined in accordance with 4.4.2.1 and for the defects specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.4 Quality conformance inspection.

4.4.1 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.4.2 Examination.

4.4.2.1 Individual. Each component shall be inspected for conformance to the dimensions covered by the Supplementary Quality Assurance Provisions (SQAP) shown on the drawings under "Major - 100 percent inspection". Nonconformance to any of these dimensions shall constitute failure of this examination and shall be cause for rejection.

4.4.2.2 Samples. Samples selected in accordance with 4.4.1 shall be examined for the major and minor defects specified in 4.5.1. AQL shall be 2.5 percent defective for major defects and 4.0 percent defective for minor defects.

4.5 Inspection procedure.

4.5.1 Examination. The components shall be examined for the following defects:

Major

101. All dimensions designated major on the SQAP other than those under 100 percent inspection not as specified.
102. Material not as specified.
103. Components missing or not as specified.
104. Workmanship not as specified.
105. Weld defects such as flux inclusions, spatter, undercutting, porosity, incorrect size, or lack of fusion with parent metal.
106. Riveting not as specified.

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Minor

201. All dimensions not designated major on the SQAP not as specified.
202. Identification marking incorrect, missing or illegible.
203. Treatment and painting not as specified.
204. Stenciling not as specified.

4.6 Inspection of preparation for delivery.

4.6.1 Quality conformance inspection of pack.

4.6.1.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.6.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.6.1.3 Examination. Samples selected in accordance with 4.6.1.2 shall be examined for the following defects. AQL shall be 2.5 percent defective.

107. Bundled in a manner and in quantities other than specified for level A.
108. Pins and retainers not in place as specified for level A.
109. Bundles and secured lifts not secured as specified for level A.
110. Strapping not as specified for level A.
111. Wood filler strips not applied as specified for level A.
112. Containers not as specified for level A.
113. Contents of containers not secured in a manner to prevent free movement for level A.
114. Marking missing, illegible, incorrect or incomplete for level A or C.

5. PREPARATION FOR DELIVERY

5.1 Packing. Packing shall be level A or C as specified.

5.1.1 Level A.

5.1.1.1 Panels (type I).

5.1.1.1.1 Deck panels (class 1). The deck panels (with pins and retainers in place) shall be packed as a bundle in quantities of two. The panels shall be shipped as a bundle or the bundled panels shall be consolidated into a

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secured lift. Each bundle shall be secured with not less than two flat steel straps conforming to QQ-S-781, type I or IV, heavy duty, class B, size as applicable. Secured lifts shall be strapped with not less than three straps specified above, with two straps applied girthwise and with one strap applied lengthwise at the center of the lift.

5.1.1.1.2 Articulator panels (class 2). The components comprising each articulator panel (with pins and retainers in place) shall be bundled together. The panel shall be shipped as a bundle or the bundles shall be consolidated into a secured lift. Each bundle and lift shall be secured as specified in 5.1.1.1.1.

5.1.1.1.3 Ramp panels; female (class 3) and male (class 4). The ramp panels of like description (with the pin and retainer in place on the female (class 3) panel) shall be packed as a bundle in quantities of two. The panels shall be shipped as a bundle or the bundles shall be consolidated into a secured lift. Each bundle and lift shall be secured as specified in 5.1.1.1.1.

5.1.1.1.4 Filler panel, deck (class 5). The panels shall be packed as a bundle in quantities of two. The panels shall be positioned top against top. The angles on each panel shall be protected against damage with a nominal 2-inch by 4-inch wood filler strip, extending the full length of each angle. The two panels shall be secured together with bolts, flat washers top and bottom, and nuts. The bolts shall go through the holes in the angles and matching drilled holes in the wood filler strip. Bolts shall be the same diameter as the holes. The panels shall be shipped as a bundle or the bundles shall be consolidated into a secured lift. Lifts shall be secured as specified in 5.1.1.1.1.

5.1.1.1.5 Filler panel, articulator and ramp (class 6). The panels shall be packed as specified in 5.1.1.1.4.

5.1.1.2 Curbs (type II).

5.1.1.2.1 Deck panel curbs (class 1). The curbs, (with the pins and retainers in place) shall be packed in a close-fitting crate conforming to PPP-C-650, style B, type III, in quantities of eight. Prior to placing in the crate the curbs shall be bundled together in the manner most convenient for packing. The bundle shall be secured as specified in 5.1.1.1.1.

5.1.1.2.2 Articulator and ramp panels curbs (class 2). The curbs (with the pins and retainers in place) shall be packed together in a close-fitting box conforming to PPP-B-601, overseas type, style optional, in quantities of 12. The contents shall be blocked, braced, and anchored within the box

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in a manner to prevent free movement. Box closure and strapping shall be in accordance with the appendix to the box specification. Strapping shall be zinc coated.

5.1.1.3 Components procured separately. Items of like description procured separately from either the panels or curbs shall be packed together as specified in 5.1.1.2.2.

5.1.2 Level C. The components shall be packed in a manner to assure carrier acceptance and safe delivery to destination at lowest ratings in compliance with Uniform Freight Classification rules or National Motor Freight Classification rules.

5.2 Marking. Boxes, bundles, or secured lifts shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The panels and curbs are intended for use in the raft section which is a component of the aluminum light tactical floating bridge.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type and class of component required (see 1.2).
(c) Time frame required for submission of the preproduction model and number of each type and class required (see 3.2).
(d) Level of packing required (see 5.1).

6.3 Preproduction model. Any changes or deviations of production components from the approved preproduction model during production will be subject to the approval of the contracting officer. Approval of the preproduction model will not relieve the supplier of his obligation to furnish components conforming to this specification.

6.4 Supplementary Quality Assurance Provisions (SQAP). The contracting officer should require the supplier to maintain records of all SQAP inspections. A suggested paragraph is as follows:

> "The supplier shall maintain complete records of all examinations and tests performed to verify the requirements of classified SQAP characteristics. The records shall include as a minimum, lot size, sample size, drawing requirement, actual measurement, number and type of deficiencies found, quantity approved, quantity rejected, corrected action taken when applicable."

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6.5 Classification changes. Changes in classification of the bridge components and assemblies between this specification and the three superseded are as follows:

Old Designation	Description	New Designation
MIL-P-52085(CE)		MIL-B-52706(ME)
Type I	Articulator	Type I, Class 2
Type II	Deck	Type I, Class 1
Type III	Ramp, Female	Type I, Class 3
Type IV	Ramp, Male	Type I, Class 4
Type V	Deck Filler	Type I, Class 5
Type VI	Articulator and Ramp Filler	Type I, Class 6
MIL-C-52084(CE)		
Type I	Deck curb	Type II, Class 1
Type II	Articulator and Ramp Curb	Type II, Class 2
MIL-P-52076(CE)		
Type I	Superstructure panel connector pin assembly	These items are included as drawings in several DL's for above components.
Type II	Superstructure curb pin assembly	
Type III	Ponton coupling pin assembly	None

6.6 Definitions.

6.6.1 Supplementary Quality Assurance Provisions (SQAP). A SQAP is a contractual requirement that supplements section 4 of the specification. SQAP indicates the minimum requirements which must be inspected on the product drawings to verify the design objectives of the product and assure interchangeability of repair parts.

Custodian:

Army - ME

Preparing Activity:

Army - ME

Project No. 5420-A208

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION MIL-B-52706(ME) Bridge, Floating: Raft Section, Light Tactical, Superstructure Components

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426 REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED

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RESEARCH, DEVELOPMENT AND ENGINEERING DIRECTORATE
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